OMNIBRIDGE POS™

Universal Rust Gateway for POS Hardware, Weighing Scales & Printers

■ NATIVE RUST AGENT ■ UNIFIED PORT 8989 ■■ ZERO BROWSER DIALOGS ■ ED25519 OFFLINE LICENSING

Connect web and cloud POS applications to physical checkout hardware via local HTTP REST endpoints. OmniBridge POS bypasses browser sandbox limitations, empowering React, Angular, Vue, PHP, Odoo, or WooCommerce systems to operate thermal receipt printers, electronic scales, and cash drawers natively, reliably, and unattended.

THE HARDWARE CHALLENGE IN WEB & CLOUD APPLICATIONS:
Modern browsers block direct access to COM serial ports and force intrusive print dialogs (`Ctrl+P`), slowing down checkout speed and requiring cashier interaction for every printed receipt.
THE OMNIBRIDGE POS SOLUTION: An ultra-lightweight native Rust daemon running in the background of each POS terminal. It exposes a unified REST API on port 8989 to dispatch thermal receipts in milliseconds, read serial scales in real time, and kick cash drawers without popups or internet dependencies.

Your Web POS / Cloud ERP	■■■■■■(Local REST HTTP :8989)■■■■■■■	OmniBridge POS (Rust)	■■ ■	Checkout Peripherals
React • Vue • Odoo • PHP • Woo	PNA / Strict CORS • JSON & PDF	Direct Spooler • Serial COM • RJ11		Receipts • Scales • Drawers • Labels

< 15 ms	< 15 MB	Auto-COM	Offline Ready
Local response latency	Ultra-low RAM usage	Serial scale auto-scan	Ed25519 crypto licensing

Supported Hardware Capabilities & Implemented Flows

Weighing Scales **RS232 / USB Serial**	Automatic candidate port discovery, buffer clearance, and real-time decimal weight extraction (kg/lb). Includes offline simulation mode for testing.
Receipt Printing **Thermal ESC/POS**	Silent unattended PDF and RAW ESC/POS job dispatch directly to the OS print spooler. Automatic Windows printer-copy name normalization.
Barcode Labels **TSPL / ZPL / RAW**	Direct RAW path for shelf labeling and barcode generation. Accepts standard thermal printer commands.
Continuous Form **Dot Matrix ESC/P**	Plain-text and ESC/P escape sequences with automatic page eject (Form Feed) for industrial dot-matrix billing printers.
Cash Drawer & Docs **RJ11 / Office PDF**	Instant drawer kick-out pulses via RJ11/RJ12 printer ports (Pin 2 and Pin 5). Full-page A4/Letter PDF printing with proportional fitting.

Quick 3-Line Integration (Web POS / Frontend)

LOCAL INTEGRATION ARCHITECTURE (Port 8989):
• **1-Click Setup:** Runs as a Windows Service, Linux Systemd daemon, or macOS Launchd agent.
• **Web Compliant:** Fully supports Chrome Private Network Access (PNA) and strict CORS origin allowlisting.
• **Built-in Dashboard:** Interactive diagnostic web panel running at http://localhost:8989/.
• **Zero Cloud Dependency:** Checkout operations execute entirely offline.

FRONTEND JAVASCRIPT CONSUMPTION:

```
// 1. Read real-time scale weight:
const { peso } = await (await fetch('http://localhost:8989/peso')).json();
// 2. Trigger cash drawer kick pulse:
await fetch('http://localhost:8989/drawer', { method: 'POST' });
// 3. Print silent thermal receipt in milliseconds:
await fetch('http://localhost:8989/ticket', {
  method: 'POST', headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ content: "RECEIPT #101\nTotal: $25.00\n" })
});
```

Core API Endpoints & Real-Time Diagnostics

Method	REST Route	Purpose & Action	Payload / Response
GET	/	Interactive visual web dashboard with 1-click test suite	HTML Dashboard
GET	/peso, /weight	Real-time physical weight read from serial scale	JSON {peso: 1.25}
POST	/ticket	Silent printing to configured thermal receipt printer	JSON / PDF / RAW
POST	/drawer	Trigger electric cash drawer kick pulse (RJ11)	Pulse RJ11 Pin 2/5
POST	/print-label	Direct RAW barcode label dispatch (TSPL/ZPL)	JSON {commands}
POST	/printers/cleanup	Auto-purge and spooler recovery for jammed queues	Spooler Recovery
GET	/health, /status	Complete health check across ports, spooler & licenses	JSON Health Status

Technical Specifications & Hardware Validation Matrix

Component	Rust Implementation	Performance / Protocol	Deployment & Validation
Core Engine	Native Rust 2021	< 15 MB RAM, sub-second	Standalone single binary
Supported OS	Cross-platform	Windows 10/11, Linux, macOS	Windows / Systemd / Launchd
Security	Offline Ed25519	Asymmetric crypto signatures	Chrome PNA & Strict CORS
Hardware	Serial & Spooler	COM auto-scan & spool heal	Pilot test recommended

Commercial Licensing & Integration Options

Per-Station / POS License
Retail Stores & Supermarkets
$35 USD / year (or $90 lifetime)

- 1-Click installer running as an automatic Windows Service.
- Per-station license (Annual or Lifetime Perpetual).
- **Local diagnostic dashboard** (:8989) with auto-healing spooler.
- Unlimited scale reading and receipt printing with zero watermarks.
- Dedicated onboarding and setup support.

OEM / White-Label Partner
Software Houses & ERP SaaS
$390 USD (One-time fee)

- Unlimited distribution under your own brand name and logo.
- Complete **Postman** collection and **OpenAPI 3.0** spec.
- Zero per-seat recurring royalties.
- Dedicated license generator tool to issue keys to your own clients.
- Ready-to-paste integration snippets for your web ERP.

Custom Engineering & Drivers
Bespoke Hardware Integrations
Bespoke Project Quote

- Custom industrial scale protocols and serial driver tuning.
- Assisted integration with Odoo, WooCommerce, PHP, React, or Python.
- Dedicated guidance during pilot deployments and field tests.
- Custom store printing workflow adaptations.
- Priority direct engineering support.

READY TO INTEGRATE HARDWARE INTO YOUR WEB POS?

Download the functional DEMO installer or request a bespoke commercial quote:

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Lead Developer: Darwin Padrino | **Official DEMO Releases:** github.com/dpadrino/omnibridge-pos-demo/releases